Lecture # 10

Thursday, 19 March 2026 11:18 am

Question 1

I have 32 bills in my wallet, in the denominations of US $ 1, 5 and 10, worth $100 in total. How many do I have of each denomination?

x y z

x y z

$\$1$ $\$5$ $\$10$

? ? ?

$$x + y + z = 32$$
$$1x + 5y + 10z = 100$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 32 \\ 1 & 5 & 10 & 100 \end{array}\right)$$

$$R_2 - R_1 \quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 32 \\ 0 & 4 & 9 & 68 \end{array}\right)$$

Can $4 \to 1$ NO

using R_1?

R_2: 0 4 9 68

$3R_1$: 3 3 3 96

-3 1 6 -28

$1 - \frac{9}{4}$

$$\frac{1}{4}R_2 \quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 32 \\ 0 & 1 & \frac{9}{4} & 17 \end{array}\right)$$

$$R_1 - R_2 \quad \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left(\begin{array}{ccc|c} 1 & 0 & -\frac{5}{4} & 15 \\ 0 & 1 & \frac{9}{4} & 17 \end{array}\right) \end{array}$$

(1) $x - \frac{5}{4}z = 15$

$x = 15 + \frac{5}{4}z$

$z = t$

$x = 15 + \frac{5}{4}t$

(2) $\Rightarrow y + \frac{9}{4}z = 17$

$y = 17 - \frac{9}{4}z$

$y = 17 - \frac{9}{4}t$

$t \in \mathbb{R}$

$\left(15 + \frac{5}{4}t,\ 17 - \frac{9}{4}t,\ t\right)$

what are possible values of t?

(x, y, z) are no. of bills.

* all are non-negative
* integer value.

$x \geq 0$ $\quad 15 + \frac{5}{4}t \geq 0$ $\quad \frac{5}{4}t \geq -15$

$y \geq 0$ $\quad t \geq -12$

$x \geq 0 \quad 15 + \frac{5}{4}t \geq 0 \quad \frac{5}{4}t \geq -15$

$y \geq 0 \quad t \geq -\frac{15}{5} \times 4 \quad t \geq -12$

$z \geq 0 \quad 17 - \frac{9}{4}t \geq 0$

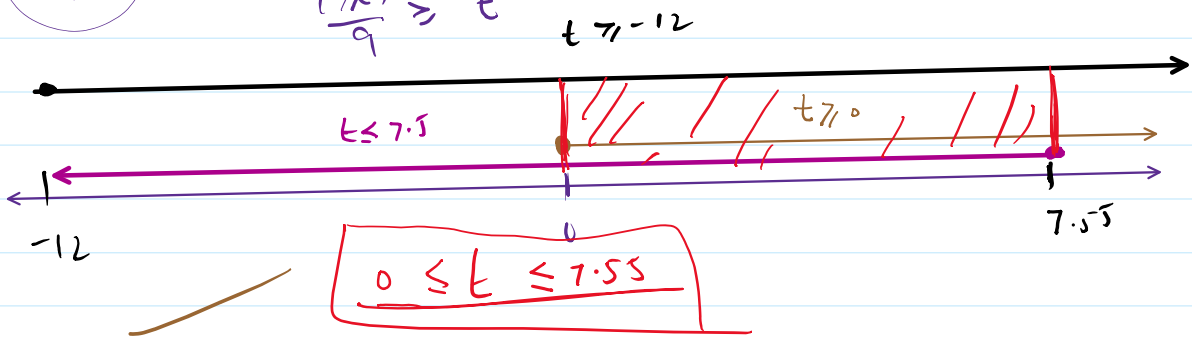
$t \geq 0 \quad 17 \geq +\frac{9}{4}t$

$\frac{17 \times 4}{9} \geq t$

$t \leq 7.55$

$\frac{2}{7}y =$

68/9=7.5556



$\boxed{0 \leq t \leq 7.55}$

$\left(15 + \frac{5}{4}t,\ 17 - \frac{9}{4}t,\ t\right)$

$t = 1;\ \left(15 + \frac{5}{4},\ 17 - \frac{9}{4},\ 1\right)$

$(16.25,\ 14.35,\ 1)$

$t = 0$

$(15,\ 17,\ 0)$ (x, y, z)

$t = 4$

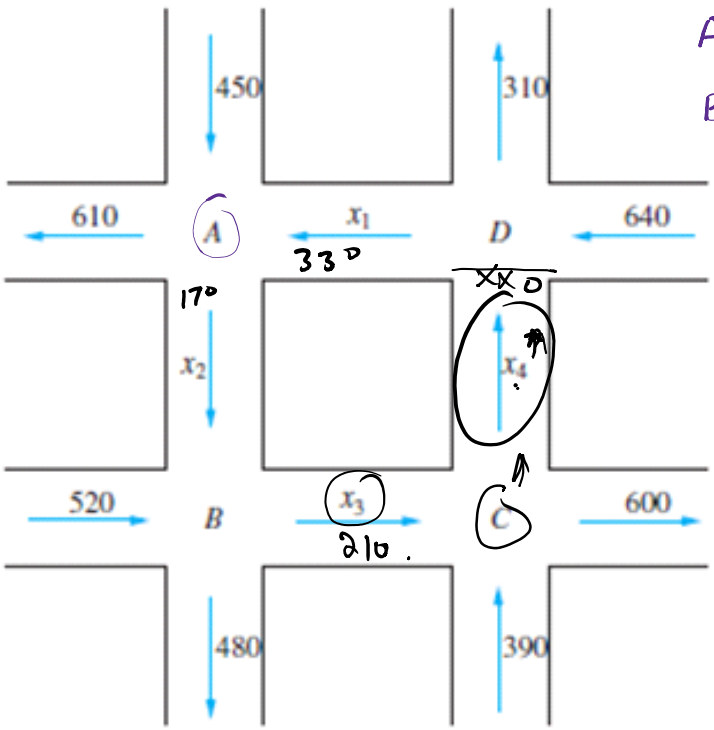
$(20,\ 8,\ 4)$

$\$1 \rightarrow 20$, 20

$\$.5 \Rightarrow 8$ 40

$\$10 \Rightarrow 4$ 40

32 $100



A: $x_1 + 450 = x_2 + 610$

B: $x_2 + 520 = x_3 + 480$

C: $x_3 + 390 = x_4 + 600$

D: $x_4 + 640 = x_1 + 310$

$x_1 - x_2 = 610 - 450$

$x_1 - x_2 = 160$

$x_2 - x_3 = -40$

$x_3 - x_4 = 210$

$-x_1 + x_4 = -330$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 160 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \\ -1 & 0 & 0 & 1 & -330 \end{array}\right)$$

$$R_4+R_1 \quad \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 160 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \\ 0 & -1 & 0 & 1 & -170 \end{array}\right)$$

$$\begin{array}{l} R_1+R_2 \\ \\ \\ R_4+R_2 \end{array} \left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 120 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \\ 0 & 0 & -1 & 1 & -210 \end{array}\right)$$

$$\begin{array}{l} \\ R_1+R_3 \\ R_2+R_3 \\ \\ R_4+R_3 \end{array} \begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \end{array} \\ \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 330 \\ 0 & 1 & 0 & 0 & 170 \\ 0 & 0 & 1 & -1 & 210 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{array}$$

$$\begin{cases} x_1 = 330 \\ x_2 = 170 \\ x_3 = 210 + t \\ x_4 = t \end{cases} \rightarrow t \geq 0, \text{ Integer}$$

$x_3 - x_4 = 210$

$x_3 = 210 + x_4$

$x_4 = t \geq 0$, Integer

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1. Introduction, linear equation.docx

2. Types of Solutions.docx

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Gauss Jordan Method.docx

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Word Problems.docx

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Friday

27-03-2026